

GAME STATUS REPORT

GAME STATUS REPORT — John Kingdom Game

Audit date: 2026-07-25 · **Branch:** `claude/shieldwall-reforged-research-k2zm28` · **Scope:** every `.rs` file in `engine/` (12,752 lines across 6 crates), design docs, and the shieldwall_reforged research corpus. Analysis only — no code changed.

All paths below are relative to `projects/john_kingdom_game/`. Line numbers are from the audited revision.

1. GAME INVENTORY — what exists so far

The project is **two games on one deterministic 120 Hz core**:

- **The battle sim** (`jk_wall` + clients `jk_bevy`, `jk_client`, harness `jk_spike`) — the Shieldwall-inspired formation game. This is the project's stated "crown jewel."
- **The TDM arena** (`jk_tdm`) — a self-contained third-person/first-person shooter side-mode with its own sim (`jk_tdm/src/sim.rs`), sharing only `jk_core` (timestep + RNG).

1.1 Implemented systems

System	Status	Where	How it works
Fixed timestep (120 Hz)	DONE	engine/crates/jk_core/src/timestep.rs:8-51	Accumulator with 8-step spiral-of-death clamp; renderer interpolation hook (<code>alpha()</code>).
Deterministic RNG	DONE	engine/crates/jk_core/src/rng.rs	Hand-rolled PCG32; separate seeded streams for spawn (<code>0x5EED</code>) and combat (<code>0xC0B47</code>) (<code>jk_wall/src/sim.rs:148,315</code>).
Calibrated constants (single source)	DONE	engine/crates/jk_core/src/constants.rs (226 lines)	Every physical number tagged <code>SOURCED / PROVISIONAL</code> per ADR-004.
Wall physics (push, crush, cohesion)	DONE	engine/crates/jk_wall/src/sim.rs:360-1016 (<code>step()</code>)	Rapier capsules + enemy-only shield colliders; lateral PD slot-holding, forward velocity-servo that saturates into othismos push; brace degradation; crush injury; α is measured, never authored.
Squad commands (6 orders)	DONE	engine/crates/jk_wall/src/command.rs:58-116	Advance/Hold/Brace/Charge/Withdraw + timed Rotate drill; each order reweights speed/spacing/push/lateral/brace/power — physics levers, not stat toggles.
Rank rotation drill	DONE	jk_wall/src/sim.rs:361-419, 500-530	Per-file stamina comparison assigns roles; diagonal pass against a captured front-plane anchor; auto-reverts after 6 s (<code>command.rs:38</code>).
Two-pool stamina	DONE	engine/crates/jk_wall/src/stamina.rs:27-43	Aerobic ceiling free; excess drains 67.5 kJ anaerobic pool; sqrt output curve to a 0.35 floor.
Melee combat (energy vs armor)	DONE	engine/crates/jk_wall/src/combat.rs:129-163, <code>sim.rs:800-878</code>	Strike = kinetic energy; armor demands energy (Williams data); penetrated J accumulates to 45 J knock-down. Block is state-based probability.
Morale / fear / rout / rally	DONE	jk_wall/src/sim.rs:895-1000, constants <code>constants.rs:131-165</code>	Fear from witnessed falls, compression, outnumbering; commander aura $\times 4$ recovery; per-man rout tolerance; rally at fear < 0.45.
Javelin volleys	DONE	jk_wall/src/sim.rs:675-798, 1154-1219	Ballistic integration, deterministic scatter, auto-volley at 14 m closure, 2 per man; player aimed throw.
Player-in-the-crush (third person)	DONE	jk_wall/src/sim.rs:469-489, <code>command.rs:119-135</code>	Velocity-servo body force-capped below crowd pressure; take-over-next-man on death.
Cohesion / breach detection	DONE	engine/crates/jk_wall/src/cohesion.rs:32-62, <code>sim.rs:1111-1137</code>	Euclidean gap $\rightarrow$ overlap ratio Ω ; breach = enemy centroid past front plane with no covering defender.
Telemetry + α fitting	DONE	engine/crates/jk_wall/src/metrics.rs	Per-tick StepMetrics; <code>fit_alpha</code> grid-search least squares.
Headless spike + battle report	DONE	engine/crates/jk_spike/src/main.rs	α sweep depth 1–8, 40v40 run, CSV/PNG/GIF/report to output/.
Scale benchmark	DONE	engine/crates/jk_spike/src/bin/bench.rs	80 $\rightarrow$ 1024 bodies; README-recorded: 80 @ 29 $\times$, 256 @ 8.7 $\times$, 500 @ 4.5 $\times$, 1024 @ 1.8 $\times$ realtime, single thread.
Battle 3D client (Bevy)	DONE	engine/crates/jk_bevy/src/main.rs (997 lines)	Procedural rigged men (or GLB drop-ins), sim-driven animation, spring camera with speed-FOV, HUD, pause menu.
Battle fallback client (macroquad)	DONE	engine/crates/jk_client/src/main.rs (338 lines)	Zero-dep renderer, same sim API.

System	Status	Where	How it works
TDM mode (whole feature set)	DONE	engine/crates/jk_tdm/src/sim.rs (2,886 lines) + main.rs (4,581 lines)	10-weapon loadout shooter: hitscan + projectiles, 4 hit zones, shield stance, dodge roll/breakfall, jump/gravity/step-up on AABB maps, 3 maps, TDM + KOTH, checkpoints, pickups, bots with 3 difficulties, full determinism test suite.
TDM client (Bevy)	DONE	engine/crates/jk_tdm/src/main.rs	First/third person, articulated chibi rigs with fingered hands, modeled weapons, viewmodel, minimap, scoreboard, settings, procedural WAV SFX.

1.2 Stubbed / referenced but not implemented

Feature	Status	Evidence
Kingdom building	MISSING	No code anywhere. Roadmap item 14 (DESIGN_MAP.md:122) marked  designed-only.
Economy (income, costs, trade)	MISSING	Fully researched (../shieldwall_reforged/research/03_economy.md — charcoal/land/transport model with proposed formulas) but zero lines of economy code .
Blacksmith / forge / metallurgy gameplay	MISSING (socket exists)	combat.rs:5-8 names the socket ("blades from the forge plug into Weapon in M4"); metallurgy constants live only in research (07_CONSTANTS.md System B). No forge code.
Raids / campaign / army management	MISSING	No code; DESIGN_MAP.md:77 "hierarchy at campaign layer  SOON ".
Terrain (battle sim)	MISSING	Flat plane only; M5 roadmap.
Arrows/archery (battle sim)	MISSING	Javelins only; "arrows (M5)" DESIGN_MAP.md:101.
Formation shapes beyond the wall	PARTIAL	Brace $\approx$ tight order; wedge/square etc.  (DESIGN_MAP.md:79).
Weapon colliders / hit locations (battle)	PARTIAL	Reach+arc query only (sim.rs:824-843); locations  SOON .
Shield durability & geometric block cone (battle)	PARTIAL	Block is probability, not geometry (combat.rs:138-151).
Slash/blunt damage split	PARTIAL	Point-attack thresholds only (ArmorKind::e_required_j, combat.rs:27-33).
Multiplayer	MISSING	Determinism groundwork exists (fixed tick, seeded RNG, bit-identical replays); no netcode.
LOD / Jolt physics swap	MISSING	ADR-002 planned trigger: ">300 bodies or client work".
glTF skeletal animation playback	PARTIAL	GLB scenes load as static drop-ins (jk_bevy/src/main.rs:295-302); no animation playback.
Save/load	MISSING	No serialization anywhere.
Casualty model from chest injury	STUB	CHEST_INJURY_FORCE_N (constants.rs:33) declared "used by the (future) casualty model" — never referenced .
breach_risk() sigmoid	STUB	cohesion.rs:13-15 defined, never called outside its file.

2. CHARACTER & UNIT SYSTEM

2.1 The battle sim: ONE archetype, procedurally varied

There are no unit *classes*. Every soldier is an Agent (engine/crates/jk_wall/src/agent.rs:32-83) whose identity is rolled at spawn (jk_wall/src/sim.rs:154-282) from the seeded PCG in deterministic order:

Stat	Value / roll	Where set
Body mass	uniform 62–82 kg	sim.rs:187, constants.rs:13
Gear mass	+ armor mass + 4.5 kg (shield 3 + spear 1.5)	sim.rs:186-189
Armor	rank 0: 35 % mail ; ranks 1+: 10 % mail ; then 50 % gambeson, remainder cloth	sim.rs:171-185, constants.rs:107-109
Weapon	70 % spear / 20 % sword / 10 % axe	sim.rs:239-246
Aerobic ceiling	300 W × uniform(0.85, 1.15) → 255–345 W	sim.rs:235
Anaerobic pool	67,500 J × uniform(0.85, 1.15) → 57.4–77.6 kJ	sim.rs:236
Strike period	weapon <code>period_mult</code> × uniform(2.4–4.0 s)	sim.rs:265-266
Initial strike cooldown	uniform(0.5–2.0 s) (staggers openings)	sim.rs:262
Javelins	2	sim.rs:263, constants.rs:118
Rout tolerance (nerve)	uniform(0.65–1.0) fear units	sim.rs:268-269
Crush tolerance	uniform(8–15 s) over 4.5 kN	sim.rs:273-275
Capsule	r 0.22 m, h 1.72 m; shield cuboid 0.9 × 1.0 × 0.2 m, massless, enemy-only collision	sim.rs:201-233
"HP"	none — 45 J of penetrated energy downs a man (<code>wounds_j</code> accumulator)	agent.rs:52, constants.rs:99

Weapon table (`engine/crates/jk_wall/src/combat.rs:68-103`):

Weapon	eff. mass	strike v	reach	period mult	pool cost	full-effort energy
Spear	0.6 kg	7.5 m/s	1.9 m	1.0 (→ mean 3.2 s)	250 J	16.9 J
Sword	0.9 kg	10.5 m/s	1.1 m	0.6 (→ mean 1.92 s)	180 J	49.6 J
Axe	1.4 kg	12.0 m/s	1.3 m	1.6 (→ mean 5.12 s)	380 J	100.8 J

Armor table (`combat.rs:27-41`): Cloth — 5 J to defeat, 0 kg; Gambeson — 30 J, 3 kg; Mail — 100 J, 11 kg.

Stats never change over time — no levels, no XP, no equipment upgrades. The only dynamic quantities are stamina pool, fear, wounds, and crush exposure. The forge (which would make weapon/armor numbers *earned*) is the missing system these structs were built to receive.

2.2 Battle-sim formulas (plain math)

- **Strike energy** (`combat.rs:107-110`): $E = \frac{1}{2} \cdot m_{\text{eff}} \cdot (v \cdot \text{clamp}(\text{effort}, 0.3, 1.2))^2$ where `effort` = `stamina_output` · (1 - 0.3·min(overload,1)) (`sim.rs:850-852`).
- **Stamina output** (`stamina.rs:40-43`): `out` = 0.35 + 0.65·√(`pool`/`pool_max`).
- **Stamina drain** (`stamina.rs:27-36`): if `P` > `aerobic`: `pool` -= (`P` - `aerobic`)·`dt`; else `pool` += 0.04 · `spare_frac` · (`pool_max` - `pool`) · `dt`.
- **Per-tick metabolic load** (`sim.rs:596-604`): `P` = 100 + `extra_order_W` + 0.9·`push_N` + 0.1·`compression_N` (that last term — `COMPRESSION_POWER_W_PER_N`, `constants.rs:58` — is *why walls rotate*: a 3 kN front press costs ~300 W).
- **Block chance** (`combat.rs:138-151`): `b` = 0.55 + 0.25·`braced` - 0.30·min(overload,2) - 0.20·(stamina<0.5), clamped [0.05, 0.95]; routing men block at flat 0.05.
- **Wound** (`combat.rs:155-162`): if $E > E_{\text{req}}(\text{armor}) \rightarrow \text{wounds} += E - E_{\text{req}}$; at `wounds` ≥ 45 J the man is down (`sim.rs:866-872`).

- **Brace degradation** (sim.rs:463-466): $\text{overload} = \max(0, (\text{compression} - 1600 \cdot \text{brace_mult}) / 1600)$; $\text{push_factor} = \text{clamp}(1 - 0.3 \cdot \text{overload}, 0.2, 1.0)$.
- **Push authority** (sim.rs:549-562): follower (not front, not charging): **250 N flat lean**; otherwise $\text{stamina_out} \cdot \text{push_factor} \cdot \text{order_push_mult} \cdot 1200 \text{ N}$.
- **Crush down** (sim.rs:880-894): $\text{compression} > 4500 \text{ N}$ accumulates exposure (decays at half rate below); down when exposure $> \text{tolerance}(8-15 \text{ s})$.
- **Fear** (sim.rs:949-999): witnessed ally down within 6 m: $+0.16 \cdot (1-d/6)$; commander falls within 12 m: $+0.30$; enemy down: $-0.06 \cdot (1-d/6)$; compression above brace limit: $+0.02/\text{s}$; outnumbering: $+0.05 \cdot \text{excess}/\text{s}$; recovery $0.015/\text{s}$ ($\times 4$ within 7 m of standing commander, $\times 3$ while routing). Rout when $\text{fear} > \text{tolerance}$; rally at $\text{fear} < 0.45$.
- **Javelin impact** (sim.rs:752-757): $E = \frac{1}{2} \cdot 0.7 \text{ kg} \cdot |v|^2$ — at the 19 m/s launch cap this lands **~126 J**, which **defeats mail** (see §3.4).
- **Cohesion** (cohesion.rs:7-9): $\Omega = \text{clamp}((0.9 - \text{gap}) / 0.9, 0, 1)$ per adjacent front pair.

2.3 TDM roster (v6)

Fighter definition: engine/crates/jk_tdm/src/sim.rs:816-859. **100 HP** (MAX_HEALTH, line 39), zone multipliers **head $\times 4.0$, arms/legs $\times 0.75$** (lines 58-60), zone bands by hit height fraction: head > 0.82 , arms > 0.66 , torso > 0.35 , else legs (apply_hit, lines 1792-1801). Movement: walk 4.8, sprint 6.6, crouch $\times 0.5$, ADS $\times 0.62$, scoped $\times 0.35$, shield $\times 0.55$, robot armor $\times 1.12$, roll 8.6 m/s (lines 32-87). Bots differ from the player only via BotParams (aim σ , reaction, engage range, aggression — lines 751-772) and a deterministic loadout rotation (lines 1086-1094). Names are cosmetic (lines 1007-1012).

Full gun table, exactly as in code (gun(), sim.rs:156-327). dmg is base torso damage/bullet; zone multipliers apply on top; projectile weapons (bow, spear) use flat damage with **no zone multiplier** (step_missiles, line 1989: "projectile damage: flat"):

Gun	class	fire s	mag	reserve	reload s	spread	spread_move	kick	dmg	pellets	projectile (v0, dmg)	zoom°
Unarmed	—	1.0	0	0	2.0	0	0	0	0	1	—	62
Glock 17	Sec	0.16	17	68	1.3	.010	.016	.0025	9.0	1	—	55
Desert Eagle	Sec	0.42	7	35	1.6	.006	.022	.008	27.0	1	—	52
MP5	Pri	0.08	30	150	1.8	.011	.015	.003	10.0	1	—	52
Remington 870	Pri	0.95	7	28	2.8	.055	.020	.010	6.5	8	—	56
AK-47	Pri	0.105	30	120	2.2	.011	.024	.0055	13.5	1	—	48
M4A1	Pri	0.09	30	120	2.0	.008	.018	.004	12.5	1	—	48
AWM	Spc	1.6	5	20	3.0	.0012	.050	.015	70.0	1	—	16 (scoped)
M249	Pri	0.075	100	200	4.5	.016	.032	.0055	11.0	1	—	52
War Bow	Spc	0.95	1	24	0.9	.004	.018	.002	34.0	1	(38 m/s, 34)	45
War Spear	Spc	1.3	1	5	1.1	.006	.015	.003	55.0	1	(17 m/s, 55)	50

Other TDM combat math:

- **Spread** (try_fire, lines 1659-1666): $\text{spread} = \text{base} + (\text{moving} ? \text{spread_move} : 0) + \text{bloom}$, then $\times 0.32 \text{ ADS}$, $\times 0.55 \text{ crouch}$; bloom grows by kick per shot ($\times 0.8 \text{ leaning}$), caps 0.05, decays 0.12/s (line 1227).

- **Shield** (lines 61-69, 1758-1782): front $\pm 60^\circ$ arc only; $\times 0.65$ damage cut standing, $\times 0.95$ crouched; nothing from sides/rear; no firing while up; drops on death/roll.
- **Robot armor** (lines 1815-1820): `absorbed = min(dmg, armor); armor -= absorbed; hp -= dmg - 0.7*absorbed` — i.e. 70 % soak until the 100-pt pool is gone.
- **Creation dynamics**: everyone spawns with a picked 3-slot loadout + shield; per-slot magazine memory (`switch_slot`, 1578-1595); respawn 3 s at base or owned checkpoint (1239-1280); spawn protection 1.2 s.

There is **no cost, training time, or upkeep anywhere in either game** — no economy exists to price anything (see §3.6).

3. BALANCE & MATH ANALYSIS

Numbers below were computed by script (`scratchpad/balance.py`) directly from the code's constants, and cross-checked against the sim's own tests.

3.1 TDM: cost vs effective power

There is no currency, so "cost" = the slot the gun occupies. TTK assumes all shots land (upper bound); STK = shots to kill.

Gun	dmg/shot	STK body	STK head	TTK body s	TTK head s	burst DPS	mag dmg	sustained DPS*
Glock 17	9	12	3	1.76	0.32	56	153	38
Desert Eagle	27	4	1	1.26	0.00	64	189	42
MP5	10	10	3	0.72	0.16	125	300	71
Rem 870 (8 pellets, point-blank)	52	2	1	0.95	0.00	55	364	39
AK-47	13.5	8	2	0.73	0.10	129	405	76
M4A1	12.5	8	2	0.63	0.09	139	375	80
AWM	70	2	1	1.60	0.00	44	350	32
M249	11	10	3	0.67	0.15	147	1100	92
War Bow	34	3	3 (no head mult)	1.90	1.90	36	34	18
War Spear	55	2	2 (no head mult)	1.30	1.30	42	55	23

* sustained = mag damage / (mag·fire_period + reload).

Findings, ranked:

1. **M249 is the best primary on every axis that matters.** Highest burst DPS (147), highest sustained DPS (92), a 100-round mag, and its recoil (`kick 0.0055`) equals the AK's. Its only taxes are base spread (0.016) and a 4.5 s reload it rarely needs. There is **no per-gun movement-speed penalty** (speed depends only on stance/ADS/shield — `sim.rs:1347-1362`), so the support gun sprints like an SMG. It is strictly better than the MP5 and dominates mid-range.
2. **No damage falloff on any hitscan gun** (`try_fire` traces to `t_hit = 200 m`, line 1705). An MP5 or shotgun pellet at 150 m does full damage; the only range tax is spread. This flattens the intended niche structure

(SMG close / rifle mid / sniper far) and makes the bow/spear objectively worse than rifles at their own long-range game.

3. **Projectiles get no headshot bonus** (flat damage, `sim.rs:1989`), while suffering travel time, drop, 1-round mags. War Bow: 3 hits $\approx$ 1.9 s TTK best case — the worst effective weapon in the game while being a "signature" pick. War Spear: 2 hits, only **5 spears total** (mag 1 + reserve 5, $\sim$ 6 kills of theoretical max). They can't compete with the AWM in the same Special slot.
4. **Glock vs Deagle is no contest.** Same slot: Deagle has better DPS (64 vs 56), a 1-tap head, 4-STK body vs 12-STK. The Glock's only edges (mag size, lower move-spread) don't buy back an 8-shot STK gap. Glock is a dead pick.
5. **Crouched shield is near-immunity from the front:** M4 needs **160** body shots through a crouched shield ($12.5 \rightarrow 0.625$ dmg), AWM needs 29. Flanking is the designed counter and roll disables it (tested, `sim.rs:2538-2551`) — defensible, but with no shield durability a corner-camper in a checkpoint ring is a stalemate the bots can't solve (bots only flank by accident of waypoints).
6. **Robot armor $\approx$ +75 % effective HP against rifles** ($8 \rightarrow 14$ M4 STK) *plus* $\times 1.12$ speed, on a 45 s respawn — and it spawns on the **same spot as the KOTH hill** (`sim.rs:1036-1039` vs `1160`): whoever holds the hill also gets the strongest item, a snowball loop in KOTH.
7. **Exhausted-overtime tie hands the win to Blue** — the player's team (`finish(Team::Blue) // blue by honor, sim.rs:1216`). Cosmetic now, but it biases every recorded stat toward Blue.
8. Doc rot: header comment `sim.rs:8-9` still says the v3 rule "5 body shots / 2 headshots"; v6 baseline is 8/2 (tested at `sim.rs:2381-2409`). Comment at `sim.rs:2458` says crouch height 1.05; the constant is 1.15 (`sim.rs:31`).

3.2 TDM difficulty scaling (sane)

Easy $\rightarrow$ Normal $\rightarrow$ Hard: aim σ .055/.030/.014, reaction .45/.28/.15 s, engage 22/35/50 m, aggression 0.6/1.0/1.4 (`sim.rs:751-772`). Monotonic, verified by a field test (`difficulty_scales_the_bots, sim.rs:2577-2606`). Note Hard bots at $\sigma=0.014$ with 0.15 s reaction out-aim the AWM's own base spread — Hard is *very* hard at range.

3.3 Battle sim: the weapon-vs-armor matrix has holes

Full-effort strike energy vs energy-to-defeat (strikes-to-down = $\text{ceil}(45 \text{ J} / \text{penetration})$; "—" = cannot penetrate ever):

	Cloth (5 J)	Gambeson (30 J)	Mail (100 J)
Spear 16.9 J	11.9 J $\rightarrow$ 4 hits	— (needs effort $1.33 > \text{max } 1.2$)	—
Sword 49.6 J	44.6 J $\rightarrow$ 2	19.6 J $\rightarrow$ 3	—
Axe 100.8 J	95.8 J $\rightarrow$ 1	70.8 J $\rightarrow$ 1	0.8 J $\rightarrow$ 57 hits ($\approx$ 5 min of unblocked swings — i.e. never)

1. **The spear — 70 % of every army — cannot wound $\sim$ 50 % of every army.** Gambeson is rolled on half the men (`sim.rs:176-185`), and $16.9 \text{ J} < 30 \text{ J}$ at any physically reachable effort (v-clamp $1.2 \rightarrow \text{max } 24.3 \text{ J}$). Force `armor_a = armor_b = Gambeson` and melee produces zero wounds; all casualties are crush. The sourced band said thrusts are 10–30 J *into gaps* — the model charges the thrust against full armor every time (no gap-finding roll).
2. **Mail is melee-proof.** Nothing in the melee triangle meaningfully opens it (the axe's 0.8 J over threshold is noise). The `mail_side_outlasts_cloth_side` test (`tests/combat_battle.rs:39-53`) passes precisely *because* of this. Fine as the metallurgy thesis — but when the forge arrives, mail is a binary win-button unless E_{req} or weapon quality gets a continuous scale (research already has one: $E_{\text{req}} = 55 \cdot Q \cdot t^{1.6} / \cos\theta$, `07_CONSTANTS.md`).

3. **The javelin is secretly the best anti-armor weapon in the game.** Impact energy = $\frac{1}{2} \cdot 0.7 \cdot |v|^2$ with $|v| \approx 19$ m/s at man-height $\rightarrow \sim 126 \text{ J} > 100 \text{ J mail threshold}$, penetrating $\sim 26 \text{ J} \rightarrow \mathbf{2 \text{ hits down a mailed man}}$ — something no melee weapon can do. The asymmetry comes from modeling: thrusts use *effective point mass* (0.6 kg), javelins use *full projectile mass* (0.7 kg) at full velocity. Physically arguable, but it inverts the expected ranged-vs-armor relationship and nobody chose it on purpose (the volley test only checks cloth).
4. **Block probability dominates melee pacing.** With base 0.55 (0.8 braced), expected unblocked spear hits on cloth $\approx 4/(1-0.55) \approx 9$ attempts $\approx \mathbf{29 \text{ s per kill}}$ — matches the slow-bleed battles in `output/BATTLE_REPORT.md`. Intentional grind, but worth knowing the knob: `BLOCK_BASE` moves overall lethality more than any weapon stat.
5. **Stamina math is coherent and validated:** pool drains in 60–90 s of hard shove (`constants.rs` test, `stamina.rs` test), strike costs (250–380 J) are negligible against the 67.5 kJ pool (~ 270 spear strikes) — stamina pressure comes from *pushing and compression*, not swinging. That's a deliberate, defensible modeling choice; it does mean "exhausted swings" effectively never happens from swinging alone.
6. Command profiles (`command.rs:58-116`) are self-consistent and every claimed property is regression-tested (charge > advance impact $\times 1.2$, brace yields least, hold recovers, rotation freshens the front — `tests/commands.rs`).

3.4 Morale numbers

Compression fear (+0.02/s) vs recovery (0.015/s) means a hard press alone *sensitizes but cannot break* a line — deliberate, documented at `constants.rs:141-145`, and the pulse-dynamics comment explains the 0.04 $\rightarrow$ 0.02 retune. Witness fear 0.16 vs rout band 0.65–1.0: $\sim 4\text{--}6$ nearby deaths in quick succession break the weakest men. Commander down = 0.30 within 12 m — nearly half the weakest man's tolerance. These ratios pass all four morale tests. One quirk: `CHEER_ENEMY_DOWN` can push fear negative within a tick before the end-of-tick clamp (`sim.rs:992`) — harmless but means cheering is slightly stronger than it looks.

3.5 Progression / upgrade curves

There are none. No blacksmith, tech, era, or level curves exist in code to audit. The only "curve"-like data is in research: bloomery yields, `E_req` quality law, charcoal economics (`07_CONSTANTS.md`, `research/03_economy.md`). Consequence: §5's redesign is a *baseline* table; the first real curve work is the forge vertical slice (§4).

3.6 Economy check

No income, no costs, no pace to verify. `research/03_economy.md` contains a complete, sourced model (charcoal = land + labor + radius; transport ratios; proposed formulas like `charcoal_output(site) = min(coppice_ha  $\times$  yield  $\times$  charcoal_frac, collier_labor_cap)`) — but not one line is implemented. Any statement about affordability would be fiction; flagged as the biggest gap between research investment and shipped code.

3.7 Unused / duplicated / contradictory stats

Item	Kind	Where
PUSH_FORCE_SUSTAINED_N (400 N)	Unused — sim uses PUSH_FORCE_BURST_N and FOLLOWER_LEAN_FORCE_N only	constants.rs:25
CHEST_INJURY_FORCE_N (1.1 kN)	Unused — crush model uses CRUSH_DOWN_FORCE_N (4.5 kN) instead	constants.rs:33
ALPHA_EXPECTED_BAND, OVERLAP_TARGET	Test/validation only (documented as such)	constants.rs:37,66
breach_risk()	Dead code — real breaches are geometric	cohesion.rs:13-15
Bow/Spear damage duplicated in <code>damage</code> and <code>projectile.1</code>	Duplication — two fields must be kept in sync by hand	sim.rs:307,322
Three gravities: bodies 18 (GRAVITY, tdm sim.rs:42), missiles 9.81×(1.0 spear / 0.55 arrow) (sim.rs:1935), battle GRAVITY_M_S2 = 9.81	Inconsistent by design ("gamey gravity") but the arrow factor 0.55 is a magic number duplicated in step_missiles (1935) and predict_arc (1876)	
v3 TTK comment "5 body / 2 head" vs v6 rule 8/2	Contradiction (doc rot)	tdm sim.rs:8-9
Crouch-height comment 1.05 vs constant 1.15	Contradiction (doc rot)	tdm sim.rs:2458 vs 31
Axe described as having "the reach to break things open"	Misleading — axe reach 1.3 m < spear 1.9 m; it's the <i>energy</i> that breaks things	jk_wall/sim.rs:238-241
Missile.id, HitEvent.at	#[allow(dead_code)] — declared API, unused by client	tdm sim.rs:919-921,942-944

3.8 Test suite

`cargo test --release` across the workspace: **all green** (see run log). Coverage is genuinely good: determinism (bit-identical replays in both sims), α -band emergence, cascade-vs-depth, all six commands' physical claims, morale (4 scenarios), TTK rules, shield arcs, checkpoints, maps validity, arc prediction ± 0.6 m. What's *not* tested: any cross-weapon balance claim (e.g. nothing asserts the M249 isn't dominant), and nothing exercises gambeson vs spear (which would have caught §3.3.1).

4. OPTIMIZATION OPPORTUNITIES

4.1 Performance (ranked by impact at target scale — 1000+ bodies)

- Unbounded telemetry growth — a real leak in live clients.** `WallSim::step()` pushes a `StepMetrics` (containing 3+ heap `Vecs`) **every tick, forever** (`jk_wall/src/sim.rs:1013-1015`). `jk_bevy/jk_client` run indefinitely: at 120 Hz that's ~430 k allocations/hour and unbounded RAM. Fix: ring buffer or a `telemetry_enabled` flag for clients. *Effort: small.*
- $O(n^2)$ scans in the wall sim.** Melee targeting scans every agent per ready striker (`sim.rs:829-842`), and the 10 Hz outnumber refresh is a full n^2 pass (`sim.rs:909-941`). At 1024 bodies that's ~1 M distance checks per refresh plus worst-case n^2 in melee. A uniform spatial grid (cell ≈ 2 m) makes both $O(n)$. This is the difference between 1.8× realtime and comfortably >5× at 1024. *Effort: medium.*
- Per-tick allocation churn.** Each `step()` allocates: follow vec, per-file sort vecs (`sim.rs:386-399`), `agent_snap` (717-724), morale `positions` (911-918), `newly_down`, plus `front_line()` building sorted vecs **6–8 times per tick** (twice for `enemy_plane` at 429-439, twice in `collect_metrics` 1077-1080, more in breach detection). Compute the front line once per tick into reusable buffers. *Effort: small-medium.*
- Rapier is single-threaded here, with 8 solver iterations** (`sim.rs:304-312`). Options in order of cheapness: enable rapier's `parallel` feature; profile 8 → 6 iterations against α -band tests; the planned Jolt swap (ADR-002 trigger ">300 bodies" is already tripped by the 1024 benchmark). *Effort: small / large.*

5. **TDM is fine at its 8v8 cap.** Linear cover scans per pellet/LOS (`ray_hit` over ~60 AABBs) and per-missile checks are negligible at this scale. Only if maps or counts grow: a BVH over `cover`. The client's `sync_fighters` does ~30 `get_mut` part lookups per fighter per frame (`tdm_main.rs:2573-2848`) — fine at 16 rigs.
6. **AI think-rate:** battle-sim behavior runs at full 120 Hz per agent (only outnumbering is throttled). `DESIGN_MAP.md:80` already plans 10–20 Hz decision throttling — the follow/role logic (`sim.rs:386-419`) is the next candidate to move to the slow tier.

4.2 Code structure — single source of truth for stats

Current state is *better than most codebases* (ADR-004 works: `jk_core::constants` + one `gun()` table + `Weapon::spear()/sword()/axe()`), but three things undercut it:

- The TDM gun table is **code**, not data. Balancing means recompiling; the in-game manual already generates from the table, so the table is one `serde` derive away from being a `weapons.ron` asset with hot-reload. Include zone multipliers, movement multipliers, and (new) falloff curves in the same file.
- Duplicates listed in §3.7 (projectile damage pairs, the 0.55 arrow-gravity factor, three gravities) should each collapse to one named constant.
- The battle sim's kit distribution (35/10 % mail, 50 % gambeson, 70/20/10 weapons — `jk_wall/sim.rs:171-246`) is buried in the spawn closure; it's the army-composition knob the campaign layer will need, so promote it into `WallSimConfig`.

4.3 Gameplay — top 10 concrete changes

#	Change	Why	Files	Effort
1	Hitscan damage falloff (e.g. $\times 1.0$ to 15 m $\rightarrow \times 0.6$ at 50 m, per-gun near/far)	Restores SMG/rifle/sniper niches; makes bow/AWM the long-range picks; fixes shotgun sniping	tdm_sim.rs try_fire/apply_hit (add distance $\rightarrow$ multiplier), gun table	Small
2	Tax the M249 : move-spread $\times 2$ while unbraced, or per-class move-speed mult (LMG 0.9)	It currently outclasses every primary (§3.1.1)	tdm_sim.rs:156-327, 1347-1362	Small
3	Projectile headshot bonus ($\times 1.5$ – 2 via hit-height on the cylinder, like hitscan) + raise spear reserve 5 $\rightarrow$ 8	Bow/spear are strictly-worse picks; they're the game's signature	tdm_sim.rs:1941-1951 (compute zone from m.pos[1]), 1989	Small
4	Give the Glock a role : 0.16 $\rightarrow$ 0.11 s fire, or 9 $\rightarrow$ 11 dmg (STK 10)	Dead pick vs Deagle in the same slot	tdm_sim.rs:187-200	Small
5	Decouple robot armor from the KOTH hill (in KOTH, spawn it on a flank pad)	Hill-holder also gets the best item — snowball (§3.1.6)	tdm_sim.rs:1034-1039	Small
6	Spear vs gambeson : add a gap-finding roll (e.g. 20 % of thrusts test vs cloth) or raise spear energy band	70 % of the army is harmless against 50 % of the army (§3.3.1)	jk_wall/combat.rs resolve_strike, constants.rs:83-96	Small-Medium
7	Unify the javelin/thrust energy model (effective mass for both, or E_req applies a projectile factor)	Javelins silently out-armor-pierce every melee weapon (§3.3.3)	jk_wall/sim.rs:752-757, constants.rs:113-129	Small
8	Cap telemetry (ring buffer, N=2 for clients / full for spike)	Unbounded RAM growth in live clients (§4.1.1)	jk_wall/metrics.rs, sim.rs:1013-1015	Small
9	Externalize the gun table to weapons.ron + hot-reload; regenerate the manual from it	Balancing without recompiles; single source of truth	tdm_sim.rs:156-327, main.rs manual page	Medium
10	The forge vertical slice — bloomery $\rightarrow$ blade quality Q $\rightarrow$ Weapon.strike_v/E_req via the researched law $E_{\text{req}} = 55 \cdot Q \cdot t^{1.6}$	It's the project's entire thesis ("Era 1: bloomery $\rightarrow$ one blade $\rightarrow$ one 40v40 wall") and every socket already exists (combat.rs:5-8)	new crate jk_forge; jk_wall/combat.rs	Large

5. RECOMMENDED STAT REDESIGN (baseline to tune)

5.1 TDM guns

Principles: M4 stays the anchor (8 body / 2 head, owner's rule). Every gun gets exactly one identity stat it wins. New columns `falloff_start/end` (damage $\times 1.0 \rightarrow \times 0.6$ linearly) and `move_mult` create the niches ranged stats can't.

Gun	dmg	fire s	STK body (close)	falloff m	move_mult	other changes	identity
Glock 17	11	0.13	10	18–40	1.0	—	fast, forgiving sidearm
Desert Eagle	27	0.42	4	15–35	1.0	kick 0.008 → 0.010	the 1-tap head, punished spam
MP5	10	0.08	10	12–30	1.05	—	close-range mobility king
Rem 870	6.5×8	0.95	2	8–20	1.0	—	door-fight monster, useless past 20 m
AK-47	13.5	0.105	8	25–55	1.0	—	hardest-hitting rifle, hardest to control
M4A1	12.5	0.09	8	25–55	1.0	—	the baseline (unchanged)
AWM	70	1.6	2	none	0.95	—	only head is instant (unchanged)
M249	11	0.075	10	20–50	0.88	spread_move .032 → .045	area denial, not a sprint gun
War Bow	34	0.95	3	none	1.0	zone mult on arrows (head ×2 → 68 = 2-hit head)	silent skill pick
War Spear	55	1.3	2	none	1.0	reserve 5 → 8 , head ×1.5 (82.5)	high-risk lob, now sustainable

Shield: keep 65/95 % but add **durability 250 dmg** (breaks, regenerates 25/s after 5 s unraised) — ends infinite turtling without touching the flanking counter. Robot armor: 100 → **75** pool, keep 70 % soak ($\approx$ +4 rifle hits instead of +6), speed bonus 1.12 → **1.06**.

5.2 Battle sim (Era-1 melee/armor)

Goal: every weapon threatens every armor *somewhere*, with the ordering preserved (axe > sword > spear vs armor; mail >> gambeson >> cloth). Uses only existing constants:

Constant	Now	Proposed	Reasoning
SPEAR_THRUST_V_MS	7.5	8.5 (E = 21.7 J)	Still inside the sourced 10–30 J band; spear now wounds gambeson on a committed thrust (pen 21.7–30 <0 — still no; see next row)
<i>new</i> GAP_FINDING_P	—	0.25	25 % of penetrating checks resolve vs one armor tier lower (thrusts hunt seams — the historical mechanism the point-data implies). With it, spear vs gambeson downs in $\sim \text{ceil}(45/16.7)=3$ gap hits; mail stays axe/javelin business
E_REQ_MAIL_J	100	100 (keep)	Sourced (Williams); axe at 100.8 J stays a coin-flip splitter — correct feel
Axe eff_mass	1.4	1.5 (E = 108 J)	Pen 8 J → mail downed in ~ 6 unblocked hits $\approx$ one sustained duel, not 57
JAVELIN_V0_MAX_MS	19	16 (E = 89.6 J)	Javelin no longer out-pierces mail; still shreds cloth/gambeson volleys
WOUND_DOWN_J	45	45 (keep)	Keeps the validated 2-hit sword / 4-hit spear vs cloth pacing
BLOCK_BASE	0.55	0.50	Battles resolve ~ 10 % faster; front-rank mail still the survivability story
Kit fractions	hardcoded	move to WallSimConfig	Campaign layer's army-composition knob (§4.2)

Sanity checks under the proposal (same formulas): spear vs cloth 3–4 hits (was 4), sword vs gambeson 3 (unchanged), axe vs mail ~ 6 unblocked (was 57), javelin vs mail stopped (was 2-hit kill), all `tests/` invariants except `mail_side_outlasts_cloth_side`'s exact $\times 3$ margin should still hold — rerun the suite after applying.

Suggested order of work — next 5 sessions

1. **Session 1 — hygiene + leaks (all small, all \$4):** telemetry ring buffer; delete/implement `PUSH_FORCE_SUSTAINED_N`, `CHEST_INJURY_FORCE_N`, `breach_risk`; collapse duplicated constants (projectile damage pair, 0.55 arrow gravity); fix the three stale comments; overtime tie → draw.
2. **Session 2 — TDM balance pass:** damage falloff + per-gun move_mult columns, M249/Glock/projectile changes, armor-off-hill, shield durability (§5.1). Add a `balance.rs` test that asserts no primary strictly dominates another ($\text{DPS} \times \text{mobility} \times \text{capacity envelope}$).
3. **Session 3 — data-driven stats:** gun table → `weapons.ron` with hot reload; battle kit fractions → `WallSimConfig`; regenerate the in-game manual from the data file.
4. **Session 4 — battle-sim combat rebalance:** §5.2 constants + gap-finding roll + javelin/thrust energy unification; add the missing gambeson-vs-spear regression test; rerun spike + morale suites.
5. **Session 5 — scale + the thesis:** spatial grid for melee/outnumber scans (target: 1024 bodies $\geq 4\times$ realtime single-thread), then start the **forge vertical slice** (`jk_forge`: bloomery yield → bar quality Q → `Weapon/ArmorKind` parameters via the researched penetration law) — the system every socket in `combat.rs` has been waiting for.

Verification artifacts: balance math script at `scratchpad/balance.py` (session temp dir); full workspace test run log referenced in §3.8.